

JETBRAINS STATE OF DEVELOPER ECOSYSTEM 2024

State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts — Insights from 23,262
developers worldwide

ENGINEERING LEADERS • CTOS • DEVELOPER EXPERIENCE TEAMS



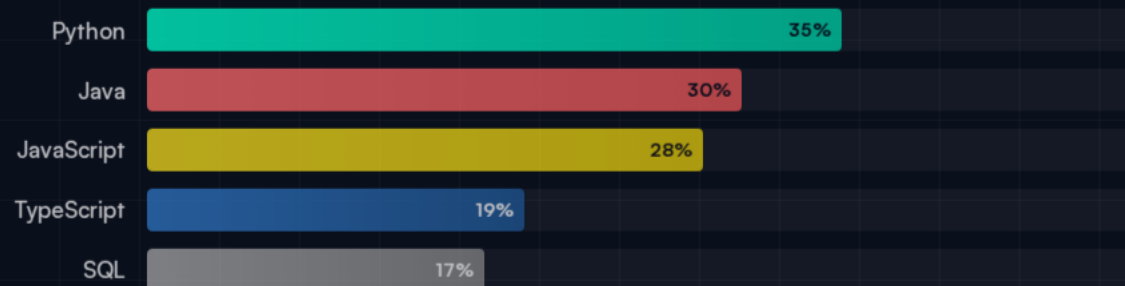
Python Closes the Gap: 8 Years of Language Usage Shifts

JavaScript peaked at 70% in 2020, but Python surged from 32% to 57%. TypeScript tripled. PHP collapsed.



Python Is Now the #1 Primary Language at 35%

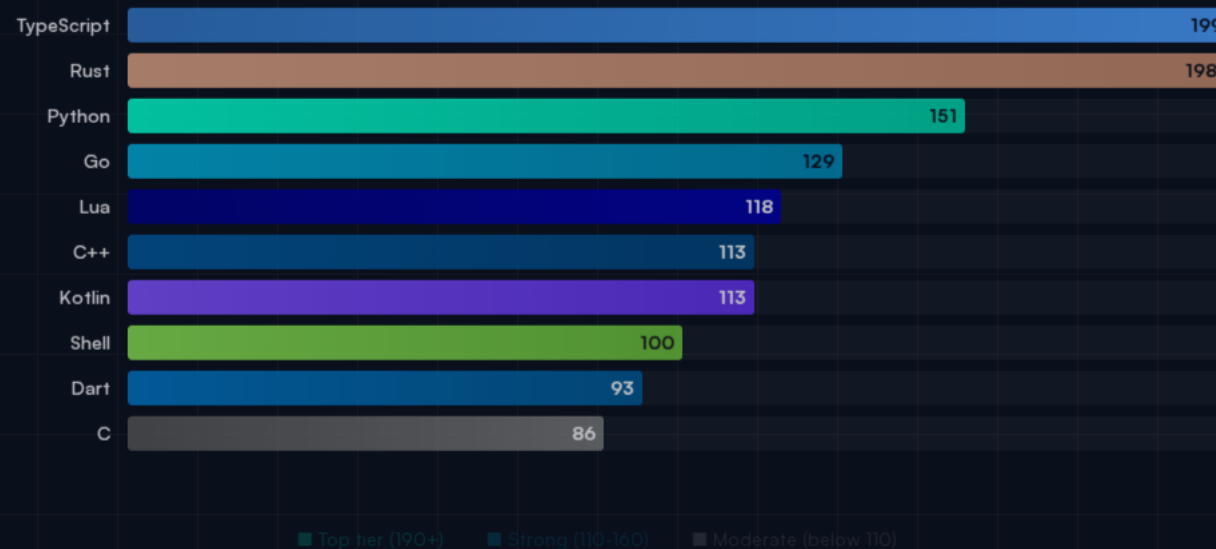
Primary language breakdown vs. adoption intent — developers rarely switch stacks.



Ecosystem Lock-In: Adoption intent for new primary languages remains below 6% across the board — only 6% plan to adopt Python, 5% TypeScript, 4% Java and SQL. Developers stick with what they know.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Type-safe languages with modern tooling dominate the top spots.



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language choice is driven by domain requirements, not general preference.

APPLICATION DOMAIN	LEADING LANGUAGE	SHARE
AI / Machine Learning	Python	31%
Data Science & Analytics	Python	28%
Integrating with APIs & Services	Python	38%
User Interface Development	TypeScript + JavaScript 58%	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%

Python dominates data-centric domains. TypeScript/JavaScript own the frontend. Java rules enterprise backends.

What This Means for Your Technology Strategy in 2025

Five actionable insights for engineering leaders and CTOs

01

Invest in Python for AI/ML

Python is the undisputed leader in AI (31%), data science (28%), and data processing. Build your AI teams around it.

02

TypeScript Is the Future of Frontend

Tripled usage to 37%, tops satisfaction at 199, and dominates UI development at 62%. Standardize on it.

03

Rust and Go Are Infrastructure Growth Plays

Both prioritize performance and concurrency with compiler safety. Ideal for microservices and cloud-native tooling.



Source: JetBrains State of Developer Ecosystem 2024 — 23,262 developers surveyed worldwide